Estimating the Impact of the Expansion of Protected Areas on Forest Loss in Colombia

A Staggered-Adoption Difference-in-Differences Design

Camilo Pedraza Jiménez, PhD Candidate, Hertie School

Abstract

Draft placeholder. The paper is scoped to protected areas only (a land-titling comparison originally planned for this project has been dropped from the main design; see Section 7). Suggested content, in order, once this is written: (1) **Question:** the causal effect of protected-area designation on annual forest loss in Colombia. (2) **Design:** a 1km-grid, 2001–2025 panel; staggered-adoption difference-in-differences, following Callaway & Sant'Anna (2021), with 19 WDPA-cohort treatment years and a 35km-buffered control group, motivated by a semivariogram-based bandwidth diagnostic. (3) **Main result:** a robust negative ATT, roughly −178 to −650 m²/cell-year (about 14–50% of the control-group mean annual loss) depending on anticipation and covariate assumptions. (4) **Methodological contribution:** a boundary-band control grid that functions as a leakage/frontier-selection diagnostic, showing near-border control choices can flip the estimated sign entirely. (5) **Contribution to the literature:** state once Section 2 is written.

Keywords: protected areas; deforestation; Colombia; staggered-adoption difference-in-differences; spatial autocorrelation

Contents

1 Introduction 3
2 Background and Literature 3
2.1 Institutional Background 3
2.2 Conceptual Channels 3
2.3 Related Literature 4
2.4 Descriptive Patterns 4
3 Data and Measurement 4
3.1 Unit of Analysis 4
3.2 Treatment Construction 4
3.3 Control Units and the Boundary-Band Design 5
3.4 Covariates 6
4 Empirical Strategy 7
4.1 Treatment Timing Notation 7
4.2 Anticipation 7
4.3 Control Group and Identification 7
4.4 Specifications 8
4.5 Standard Errors and Inference 8
5 Results 9
5.1 Baseline Estimates 9
5.2 Dynamic Effects 10
5.3 Overall ATT by Control Band 11
5.4 Heterogeneity 12
6 Mechanisms and Robustness 14
6.1 Mechanisms 14
6.2 Bandwidth Sensitivity 14
6.3 Robustness Checks 17
6.4 Threats to Interpretation 17
7 Conclusion 17
8 Appendix 18
8.1 Notation Summary 18
8.2 Data Construction Details 18
8.3 Additional Figures and Tables 18
References 19

1 Introduction

Placeholder — the paper's scope is now protected areas only. Suggested structure, with concrete anchors already available from the completed design (Section 3, Section 4, Section 5):

1. **Substantive question.** Does formal protected-area designation reduce deforestation, and how sensitive is that estimate to where the control group is drawn relative to a PA's border?
2. **Why it matters.** Colombia has expanded its PA network substantially since 2001 (19 designation cohorts, 103,488 treated 1km cells in the analysis sample); whether that expansion actually reduces loss, rather than just displacing it, is a first-order question for both conservation policy and the broader PA-effectiveness literature.
3. **Identification challenge, previewed.** Two threats dominate: (a) staggered adoption, handled with the group-time estimator of Callaway & Sant'Anna (2021) rather than a naive two-way fixed-effects regression; and (b) the control group itself is not neutral — forest-loss pressure and the raw sign of the estimated effect both change systematically with distance to a PA border (Section 6.2), which could reflect genuine leakage, frontier-selection siting, or both.
4. **Main findings and contribution.** Preview the headline range of ATT estimates (Table 4) and the boundary-band diagnostic's finding that near-border estimates flip sign under covariate adjustment (Figure 4) — this second point is a methodological contribution in its own right, independent of the headline number.

Useful things to reference directly here once drafted: Figure 2 (treated/control geography), Table 4 (headline ATT by specification), and the semivariogram range (Figure 7) that motivates the 35km buffer.

2 Background and Literature

Placeholder, refocused on protected areas only (the Afro/Indigenous/Campesino titling institutional background is no longer part of this paper's design and can be cut or moved to a future paper).

2.1 Institutional Background

Suggested content: how protected areas are designated in Colombia (designation authority, IUCN categories present in the sample — e.g. the Regional Integrated Management District category behind the cohort-2013 case in Section 5.4); the pace and geography of PA expansion since 2001 (19 cohorts, no new cohort in 2015–2017 or 2022–2023 — worth a sentence on why, if known); why designation timing is plausibly non-random (frontier siting, conflict/state-presence history), previewing Section 6′s frontier-selection discussion.

2.2 Conceptual Channels

Suggested channels, now that the design is PA-only: restrictions on land use and enforcement presence inside a PA; anticipatory behavior before formal recognition (motivating the anticipation specification in Section 4); displacement/leakage to adjacent unprotected land (the mechanism the boundary-band diagnostic in Section 6.2 is built to probe); frontier selection, i.e. PAs sited at the edge of accessibility frontiers (Andam et al., 2008; Joppa & Pfaff, 2009), which would produce the same near-border gradient as leakage without any causal effect of protection.

2.3 Related Literature

Suggested buckets: protected areas and deforestation / matching-based PA-effectiveness estimates ((Andam et al., 2008), (Joppa & Pfaff, 2009)); staggered-adoption causal inference ((Callaway & Sant'Anna, 2021)) and doubly-robust estimation ((Sant'Anna & Zhao, 2020)); spatial dependence and inference in observational/quasi-experimental designs ((Conley, 1999)); remote-sensing-based gridded forest-loss measurement ((Hansen et al., 2013)).

End this section by stating the gap: most PA-effectiveness designs do not treat the choice of control-group bandwidth as a testable object in its own right. This paper's boundary-band grid (Section 6.2) is positioned as addressing that gap directly, alongside the headline ATT estimate.

2.4 Descriptive Patterns

Where deforestation pressure sits, cross-sectionally. Using the annual forest-loss measure described in Section 3, Figure 1 plots raw mean annual loss as a function of signed distance to the nearest PA's true polygon boundary (negative = inside, positive = outside), pooling both treated and never-treated cells. Loss rises steadily from a low, flat level deep inside a park to about 1,240 m^2/year right at the edge, then jumps sharply to roughly 1,700 m^2/year just outside, stabilizing across the exterior range. This is a purely descriptive snapshot (it mixes pre- and post-treatment years for treated cells), but it establishes that the PA border is a real structural discontinuity in the landscape, not an arbitrary line.

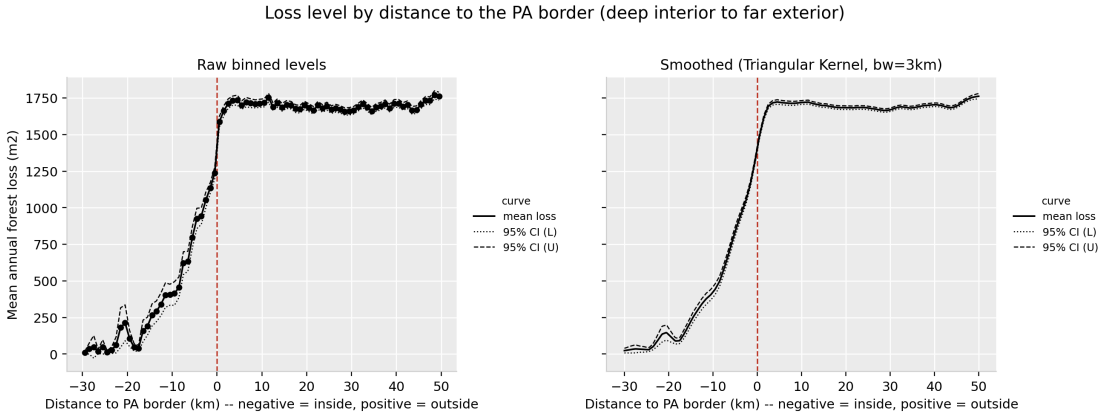


Figure 1: Loss level by distance to nearest PA boundary (deep interior to far exterior)

3 Data and Measurement

3.1 Unit of Analysis

The unit of analysis is a 1km×1km grid cell by year panel covering continental Colombia, running from 2001 to 2025. The outcome variable, annual forest loss (in square meters per cell-year), is constructed from the Hansen Global Forest Change product (Hansen et al., 2013), the same source used both for the outcome and for the baseline-forested-area covariate described in Section 3.4.

3.2 Treatment Construction

The treatment group is every 1km×1km grid cell that is added to a protected area (PA) for the first time during the study window. PA boundaries and designation years come from the World Database on Protected Areas (WDPA) (UNEP-WCMC and IUCN, 2024), pulled as

yearly snapshots so a cell's treatment cohort reflects the designation year on record at that snapshot rather than the present-day boundary. The panel begins in 2001; treatment cohorts span 2002–2025 (no new cohort is observed in 2015–2017 or 2022–2023), giving 19 annual cohorts and 103,488 treated cells in total. This treated set is held fixed across every specification and robustness check in this paper since it is not restricted to cells near a PA border. In this sense, the treatment effect being estimated is the effect of bringing a cell under protected-area status on its annual forest loss, averaged over the entire protected population rather than just cells at a PA's edge.

Cells that were already inside a PA before 2001 ("baseline" protection) are excluded from the regression sample entirely: they fail the treatment-cohort condition (their designation year does not exceed the panel start) and they are not eligible as controls either (they are not never-treated). They appear on the map in Figure 2 purely for descriptive context, to make clear that the study-period cohorts expand an existing PA network rather than starting from zero.

3.3 Control Units and the Boundary-Band Design

The never-treated cell pool is not homogeneous: forest-loss pressure varies systematically with proximity to a PA (documented in Section 6.2), so which never-treated cells are admissible as "clean" controls is itself a design choice.

- **Main buffer** (Section 5): controls are never-treated cells whose distance to the nearest treated cell stays above a single 35km threshold for the entire panel. This threshold was established after the semivariogram evidence in Section 6.2 showed that spatial correlation in the estimator's own influence function does not reach its floor until roughly that distance.
- **Boundary-band robustness design** (Section 6.2): controls for this robustness check are instead defined by a two-sided band of distance to the nearest already-designated PA: to its true polygon boundary for the near-border bands, and to the nearest treated cell for the band surrounding the main buffer directly. Six bands were defined for a reasonable comparison ([2,10], [2,15], [5,15], [5,25], [10,25], [25,35] km), plus the unbounded >35 km ring that is exactly the revised main specification's control threshold. Regarding the lower bound, a 2km minimum was chosen to avoid including cells that may be inside a PA but misclassified as outside due to the 1km grid's coarse resolution.

Table 1 shows the number of control cells available under each band (100,153 to 366,222 cells; treated cells are constant by construction at 103,488). The ordering is not monotonic in distance: >35km (264,952 cells) is smaller than the much narrower 5–25km ring (366,222 cells). This happens since the control pool size does not depend on the radius alone: a near-border band only requires a cell's distance to its single nearest PA to fall in a window, which much of the country satisfies given how many separate protected areas exist. However, the unbounded far ring requires being simultaneously far from every treated cell nationwide, a strictly harder condition once hundreds of protected areas are scattered across the landscape. Figure 2 maps the geography of four bands (2–10km, 10–25km, 25–35km, and the unbounded >35km population) against the treated grid cells and the pre-2001 baseline PAs.

Control band	n control cells
2–10km	218,251
2–15km	318,954
5–15km	227,652
5–25km	366,222
10–25km	239,273
25–35km	100,153
>35km (main buffer)	264,952

Table 1: Control-group size by boundary band

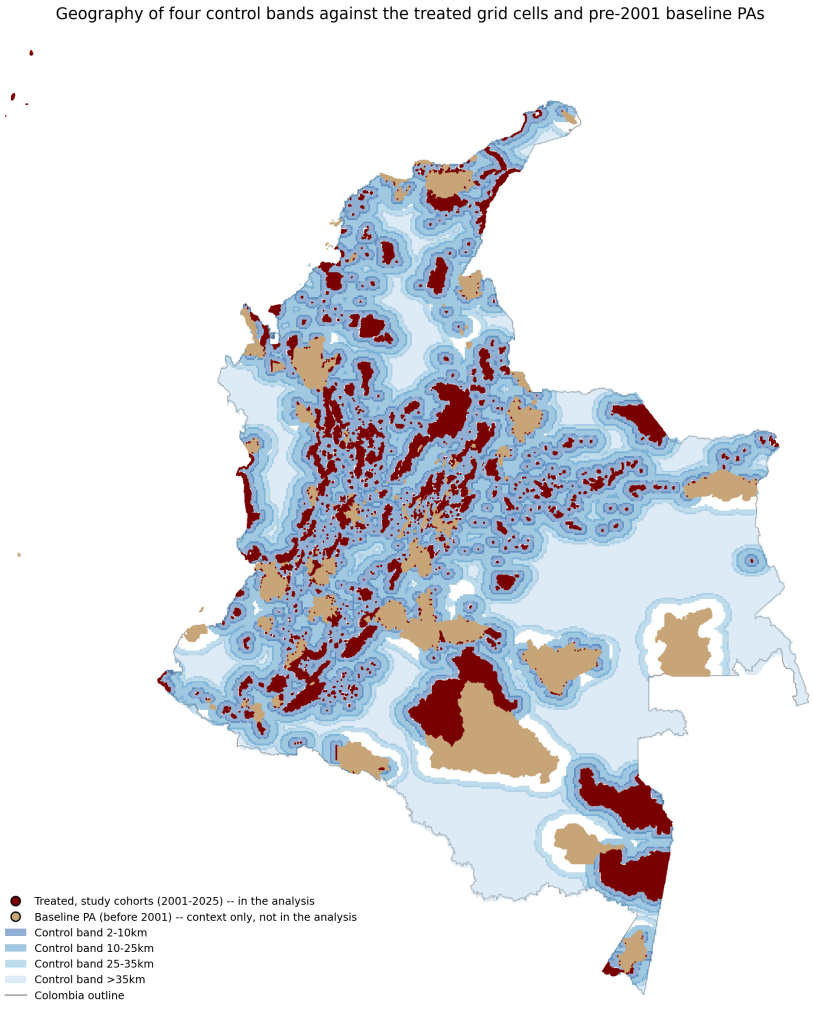


Figure 2: Geography of four control bands against the treated grid cells and pre-2001 baseline PAs

3.4 Covariates

The doubly-robust specifications in Section 4 adjust for eight covariates: elevation, slope, and terrain ruggedness (derived in Earth Engine from the SRTM GL1 digital elevation model (Farr et al., 2007)), baseline forested area as of 2000 (the same Hansen Global Forest Change source (Hansen et al., 2013) behind the forest-loss outcome variable for this research), pre-treatment DMSP-OLS and VIIRS nighttime-lights intensity (Baugh et al., 2010; Elvidge et al., 2017), and pre-treatment coca-cultivation intensity (UNODC – Sistema Integrado de Monitoreo de

Cultivos Ilícitos (SIMCI), 2023), together with an explicit missingness indicator. For two of these, an imputation or missingness-handling decision was required, and the mechanism is documented here:

- **Nighttime lights**: DMSP-OLS is essentially complete until 2013, but VIIRS is missing for 10.4% of the analyzed sample, not at random. The VIIRS sensor only begins producing usable composites around 2012–2013, so VIIRS is missing for 100% of cells in every cohort designated 2002–2012 and 0% of cells in cohorts designated 2013 or later. Therefore, VIIRS is regression-imputed from DMSP (which covers the missing years) plus the geography covariates, fit by ordinary least squares on the cells where VIIRS is genuinely observed ($R^2 = 0.380$) and applied to the 48,778 cells where it is missing. Because the imputed values still vary with DMSP and geography cell by cell, it supplies real identifying variation for the early cohorts.
- **Coca-cultivation intensity**: this variable is the one candidate that is genuinely missing for most of the sample, since coca crops are present in specific areas of the country. Given existing evidence of a strong positive relationship between coca cropping and deforestation (Marín Llanes et al., 2024), it is worth keeping this variable in the analysis. Therefore, missing values are zero-filled and a "coca missing" indicator is added as a dummy variable indicating the absence of coca cropping rather than a source of sample loss.

4 Empirical Strategy

4.1 Treatment Timing Notation

Let $g \in \{2002, ..., 2025\}$ denote cell i's first year of protected-area designation (its treatment cohort), with $g = 0$ for never-treated cells. The panel is unbalanced in treatment timing but balanced in calendar time, running $t \in \{2001, ..., 2025\}$. Because adoption is staggered and absorbing, the design follows the staggered-adoption difference-in-differences framework of Callaway & Sant'Anna (2021), built around group-time average treatment effects

$$\text{ATT}(g, t) = \mathbb{E}[Y_t(g) - Y_t(0) \mid G_i = g],$$

the average effect of treatment, for cohort g, measured at calendar time t, relative to each cell's own untreated potential outcome.

4.2 Anticipation

The `anticipation` parameter shifts the reference (pre-treatment base) period used for every $\text{ATT}(g, t)$ comparison one year earlier, from $g - 1$ to $g - 2$. Standard staggered-adoption estimators assume no anticipation: that treatment has zero effect before its recorded start date. In this setting, formal PA designation is typically preceded by public announcement, boundary demarcation, and early enforcement presence, so land users may already change behavior in the year immediately before the recorded designation year. Setting anticipation= 1 avoids folding that anticipatory year into the "clean" pre-period baseline. Mechanically, this also means every post-period estimate is measured against a base period one year earlier than in the anticipation $= 0$ version, so it integrates one additional year of treated-control divergence and will tend to be larger in magnitude even absent a true anticipation effect.

4.3 Control Group and Identification

The control group for the main specification is the single-sided main buffer described in Section 3: never-treated cells whose distance to the nearest treated cell stays above 35km for the entire panel. The core identification threat this buffer addresses is that forest-loss pressure is

not uniform in distance to a PA border (Section 6.2): a control cell drawn too close to a treated cell's border risks either genuine leakage (displaced clearing activity) or frontier-selection bias (PAs are non-randomly sited at the edge of accessibility frontiers), both of which would make a near-border "control" systematically different from the treated population even absent any causal effect of protection.

4.4 Specifications

Four specifications are estimated, crossing two covariate choices with the two anticipation assumptions above:

Label	Anticipation	Covariates
`baseline`	0	none
`anticipation1`	1	none
`dr_geography`	0	8 covariates (geography + nighttime lights + coca), doubly robust
`dr_geography_anticipation1`	1	8 covariates (geography + nighttime lights + coca), doubly robust

Table 2: Specifications estimated throughout the paper.

The no-covariate specifications (`baseline`, `anticipation1`) use none of these covariates, while the doubly-robust specifications (`dr_geography`, `dr_geography_anticipation1`) use all eight together. The doubly-robust estimator of Sant'Anna & Zhao (2020) is implemented in the staggered difference in difference framework of Callaway & Sant'Anna (2021).

The no-covariate specifications are estimated with the `csdid` Python package (Callaway et al., 2024), while the doubly-robust specifications are instead estimated with an "in-house" doubly-robust plug-in estimator that allowed for a more flexible treatment of the eight covariates for the IPW estimation (regularization, cross-validation, and trimming). The two estimation tools are therefore not used interchangeably across specifications: `csdid` is only used for the no-covariate specifications, while the doubly-robust plug-in is only used for the doubly-robust specifications.

4.5 Standard Errors and Inference

Standard errors, by contrast, are computed in-house for all four specifications using the same Conley-style spatial-HAC (heteroskedasticity-and-autocorrelation-consistent) correction (Conley, 1999), which allows nearby cells' per-unit influence contributions ψ_i to be spatially correlated:

$$\widehat{\mathrm{Var}}(\hat{\tau}) = \sum_i \psi_i^2 + \sum_{i \neq j,\, d_{ij} \leq 35 \text{ km}} K\big(d_{ij} \div 35 \text{ km}\big)\, \psi_i \psi_j$$

using a Bartlett kernel $K(\cdot)$, a 35km cutoff, and a streamed KD-tree search so the full pairwise distance matrix is never materialized. The no-covariate specifications use a plain 2×2 diff-in-means ψ_i; the doubly-robust specifications use the doubly-robust plug-in ψ_i following Sant'Anna & Zhao (2020), which treats the fitted nuisance functions as known and so omits their Theorem 1 asymptotic correction terms. `csdid` does supply its own analytic SE as a byproduct for the no-covariate specifications, but the analysis does not consider this type of standard error. The one exception is Table 3, which compares `baseline`'s analytic and spatial-HAC SEs directly to show how much the spatial correction alone changes inference. When looking at the RMS post-treatment SEs, the two are close for the main buffer once the buffer and cutoff were widened

to clear the correlation floor identified in Section 6.2. Thus, the residual spatial correlation left in the estimator's own errors barely matters here.

SE convention	Assumption	RMS post-treatment SE[1]
Analytic (csdid)	independence	25.89
Spatial-HAC	allows spatial correlation	25.94

Table 3: Analytic vs. spatial-HAC standard error for the `baseline` specification

5 Results

5.1 Baseline Estimates

Specification	ATT (m²/cell-year)	ATT (% of control mean)[2]	Significance[3]
`baseline`	−178.3	−13.6%	***
`anticipation1`	−339.4	−26.0%	***
`dr_geography`	−233.9	−17.9%	***
`dr_geography_anticipation1`	−650.4	−49.8%	***

Table 4: Overall ATT by specification for the main 35km-buffer sample

The no-covariate specifications point to a robust, negative effect of protection on annual forest loss: `baseline` estimates a reduction of about 178 m^2 per treated cell-year, rising to 339 m^2 under `anticipation1`, consistent with the mechanical amplification described above. Both are estimated with high significance under the spatial-HAC correction. The doubly-robust specifications, estimated with the per-cohort-pooled, regularized, trimmed plug-in described in Section 4, produced the following estimates: `dr_geography` estimates −233.9 and `dr_geography_anticipation1` estimates −650.4. Both have the same sign as their no-covariate counterparts, and are significantly estimated.

In percentage terms, `baseline` indicates that bringing a cell under protected-area status reduces its annual forest loss by 13.6%, relative to the rate observed on comparable unprotected land, rising to 26.0% under `anticipation1`. The doubly-robust specifications are larger in magnitude: `dr_geography` indicates a 17.9% reduction, and `dr_geography_anticipation1` indicates a nearly 50% reduction, both relative to the same control mean. The anticipation assumption matches the expected direction: the ATT is larger in magnitude when the pre-period baseline is shifted one year earlier, consistent with anticipatory behavior mentioned in the previous section.

[1] RMS-combined across post-treatment ATT(g,t) cells as $\sqrt{\sum_i \mathrm{SE}_i^2}/n$.

[2] Control mean is the mean annual loss among main-buffer control cells, pooled across the full panel (1306.5 m^2/cell-year). Each percentage is the ATT divided by this control mean, times 100.

[3] Significance is based on the spatial-HAC standard error, with $p < 0.001$ (***), $p < 0.01$ (**), $p < 0.05$ (*), and $p \geq 0.05$ (n.s.).

5.2 Dynamic Effects

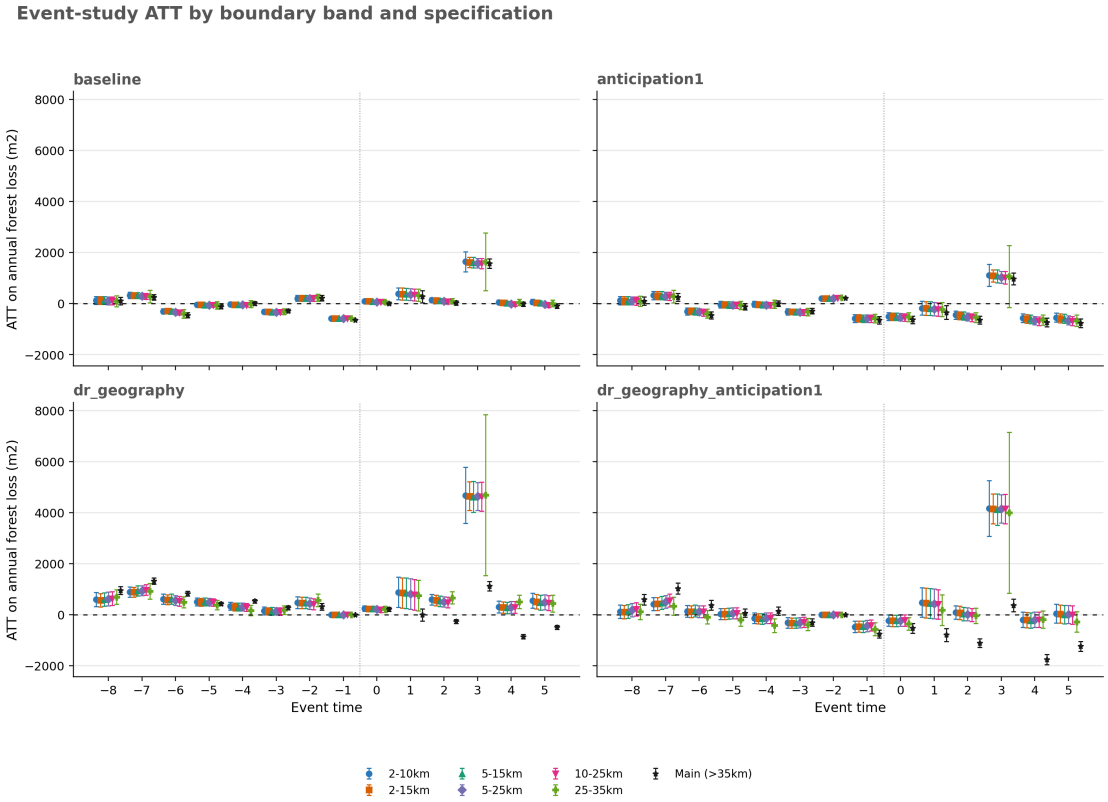


Figure 3: Event-study ATT by control boundary band and specification.

Figure 3 shows the event-study ATT for all four specifications by boundary band, rather than the single overall ATT. Pre-treatment event times ($t < 0$) are nearly indistinguishable across bands within a given specification. These pre-trend estimates oscillate within a few hundred m^2 of zero without a clear one-directional drift, which can serve as visual evidence for the parallel-trends assumption that underlies the staggered-adoption design.

When looking at post-treatment ($t \geq 0$) effects, the near-border bands tend to sit visibly higher (more positive, or less negative) than the far bands at most event times, for all four specifications. `baseline` is mostly non-significant across every band and event time. `anticipation1`, by contrast, turns clearly negative, consistent with its larger anticipation-adjusted overall ATT mentioned before. The doubly-robust specifications show the opposite pattern near the border: `dr_geography` and `dr_geography_anticipation1` are positive or non-significant across the near-border bands and only turn negative once the control band reaches the clean main buffer, which is consistent with the sign-flip-with-distance pattern already described for the overall ATT. Standard errors are also visibly smaller at the main buffer than at any near-border band, and including covariates further reduces the near-border bands' sensitivity to outlier cohorts, such as the cohort-2013 spike at post-treatment event time 3 (Section 5.4).

5.3 Overall ATT by Control Band

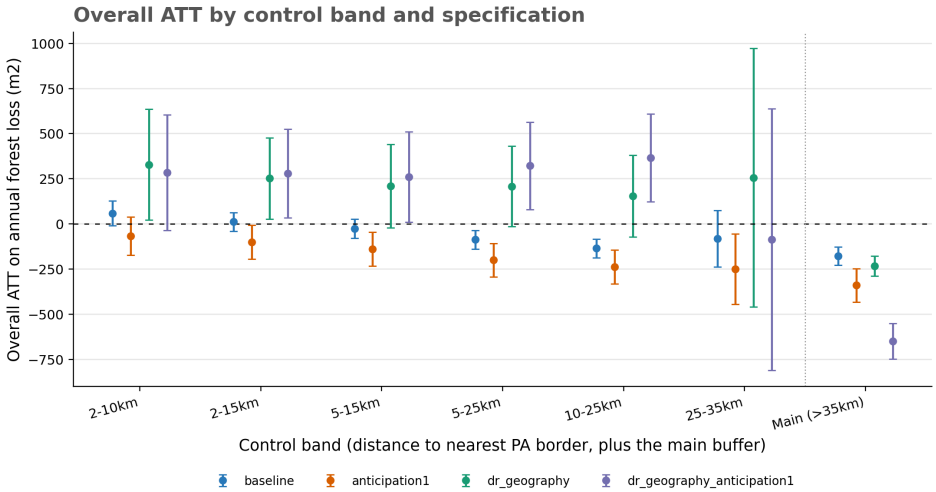


Figure 4: Overall ATT by control band and specification, boundary-band robustness grid, plus the main $>$ 35km buffer as a reference series.

Figure 4 shows the estimates for all four specifications against the two-sided boundary-band control definitions introduced in Section 6.2, holding the treated set fixed throughout, and adds the main >35km buffer as a reference. The no-covariate specifications show the gradient already documented for `baseline` alone in Table 6 from Section 6.2: the point estimate moves from small and only marginally significant near the border (2–10km: `baseline` +58.6, $p = 0.093$) to increasingly negative and significant farther out (10–25km: −135.2, $p < 0.001$), landing on −178.3 in the main buffer. This finding suggests that the control band's distance to the border matters in the expected direction.

The doubly-robust specifications tell an additional story across every near-border band. From 2–10km through 25–35km, both `dr_geography` and `dr_geography_anticipation1` are mostly positive, showing the opposite sign from the no-covariate specifications in the same bands. Additionally, the confidence intervals for the near-border bands are much wider than the ones for the main buffer. Only once the control pool moves to the clean main buffer do the doubly-robust point estimates flip to match the no-covariate specifications' sign and lead to more robust confidence intervals.

The previous finding matches the mechanism discussed in Section 6: near the border, the covariates the doubly-robust estimator relies on to adjust for confounding (for instance, night-lights) carry a different, locally reversed relationship to treatment than they do over the full sample, while the raw near-border signal is itself leakage-contaminated. Therefore, the covariate adjustment there is not cleanly removing confounding, but reacting to a less stable local relationship. The boundary-band grid is therefore best read as a **leakage diagnostic**: a near-border comparison can flip sign once adjusted for covariates, which is itself evidence against trusting it. The main 35km buffer, where the covariate-adjusted and unadjusted estimates agree, is the specification worth considering for robust results.

5.4 Heterogeneity

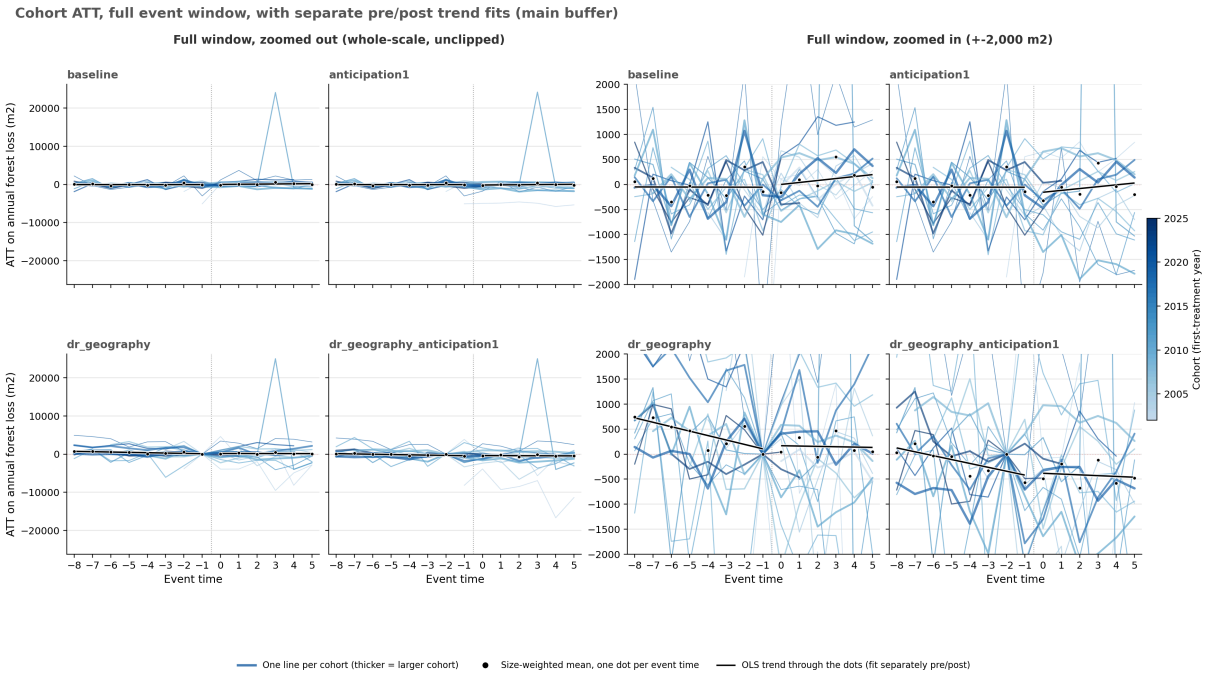


Figure 5: Cohort ATT, full event window, with separate pre/post trend fits (main buffer)

The event-study panels above pool every cohort together at each event time, which can hide cohort-specific behavior behind a simple average. Figure 5 plots each of the 19 cohorts' own $\text{ATT}(g,t)$ trajectory directly across the full event window, main buffer, all four specifications. Each line represents a cohort, with darker colors indicating later cohort years and line width encoding cohort size. The cohort-size-weighted mean is shown as a plain dot at each event time, with an ordinary-least-squares trend fit separately to the pre-period and post-period dots, so the pre-trend and post-treatment slopes can be compared directly.

The left panel, with its whole-scale, unclipped view, shows the full range of cohort-event estimates, making the outlier at post-treatment event time 3 for cohort 2013 clearly visible. The right panel zooms in to ± 2000 m^2 for a closer read of the cohorts' trajectories. The pre-period fit stays close to zero for all specifications, while the post-period fit sits mostly below zero, mirroring the negative overall ATT in Table 4. Individual cohorts oscillate around zero in the pre-period without a clear one-directional drift, and so do the weighted means – consistent with the parallel-trends assumption.

Table 5 reports each cohort's share of the cohort-size weighting separately for the pre-period ($t \leq -1$) and post-period ($t \geq 0$) weighted means. Cohort size is fixed, but each cohort's influence on the weighted mean still varies by event time, since not every cohort has an estimate at every event time. Cohort 2002, for instance, was designated one year after the panel start, so 2001 (its only possible pre-treatment year) is also its own base period $g-1$, which by construction is never assigned its own $\text{ATT}(g,t)$ estimate. It therefore contributes zero pre-period estimates despite being a moderately large cohort (11,294 treated cells), so its weight in the table below is exactly 0.0% pre-treatment versus 10.9% post-treatment. The set of cohorts behind a pre-period weighted mean is therefore not automatically the same set, with the same relative weights, behind a post-treatment one.

Cohort	n cells	Share of weight (pre)	Share of weight (post)
2002	11,294	0.0%	10.9%
2003	93	0.1%	0.1%
2004	41	0.0%	0.0%
2005	1,104	1.2%	1.1%
2006	773	0.8%	0.7%
2007	4,207	4.6%	4.1%
2008	2,662	2.9%	2.6%
2009	14,394	15.6%	13.9%
2010	1,118	1.2%	1.1%
2011	12,195	13.2%	11.8%
2012	897	1.0%	0.9%
2013	1,730	1.9%	1.7%
2014	368	0.4%	0.4%
2018	11,890	12.9%	11.5%
2019	9	0.0%	0.0%
2020	24,879	27.0%	24.0%
2021	2,119	2.3%	2.0%
2024	10,865	11.8%	10.5%
2025	2,850	3.1%	2.8%

Table 5: Approximate cohort weight in the size-weighted mean, by pre- and post-treatment window

Outlier diagnostic: cohort 2013. The previous section allowed the detection of a clear outlier: cohort 2013 at event time 3 (calendar year 2016) reports ATT over 20,000 m² across all four specifications, far higher than any other (cohort, event-time) combination. Cohort 2013 carries only 1.7% of the weight in the cohort-weighted overall ATT (Table 5), but the outlier is large enough to visibly distort simple, unweighted per-event-time averages, including the pooled event-study panels in Figure 3 and the within-cohort figures above.

Figure 6 traces this spike to its source rather than leaving it as an unexplained number. Of the cohort's 1,730 cells, only 300 have any recorded loss in 2016 at all (the median is zero), and only 106 of these cells lost more than 100,000 m² that year. Those 106 alone account for 93.3% of the cohort's total 2016 loss. Those 106 "driver" cells can be found inside and around the **Lago Azul Los Manatíes** PA, whose designation of 2013 matches this analyzed cohort. Splitting the cohort's raw annual loss trajectory into driver and non-driver cells shows that the non-driver cells follow the trend of the never-treated control pool closely throughout 2001–2025, while the spike is driven when adding the driver subset.

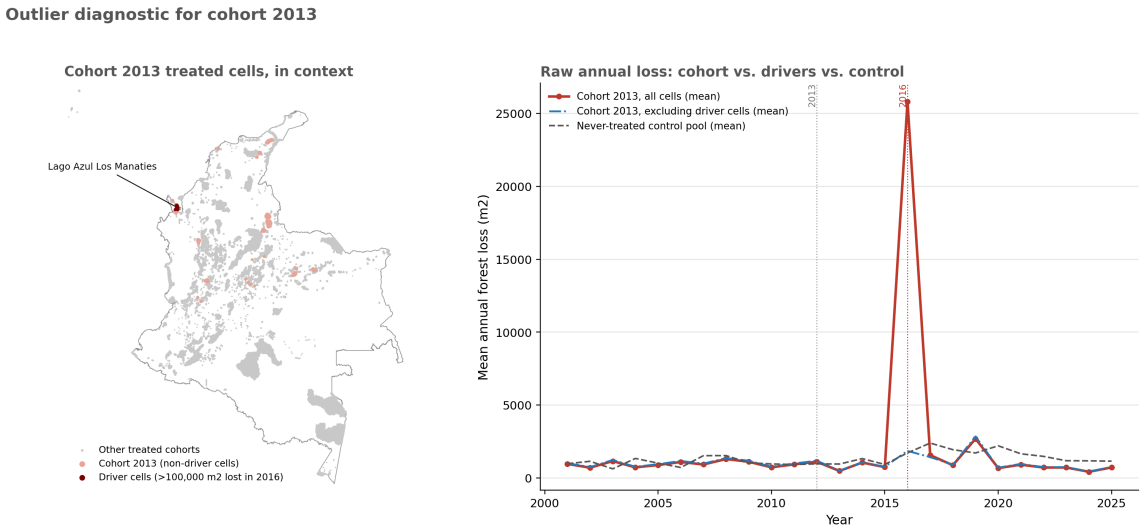


Figure 6: Outlier diagnostic for cohort 2013

This location and timing match a well-documented, independently reported event rather than a data anomaly: Colombia's national deforestation statistics for 2016 show a 44% increase over 2015, and specifically attribute the single largest fire-driven deforestation area detected in the country since 1992 to a March 2016 fire on the Antioquia–Chocó border, amid a wider wave of forest fires in Antioquia during that year's El Niño-driven dry season (Revista Semana, 2017). The Antioquia/Chocó border location, home to Lago Azul Los Manatíes, and the year (2016) match precisely the location and timing of the driver cells identified above. The spike is therefore read as a real, externally corroborated clearing event concentrated in a small number of cells and not a product of the estimation.

6 Mechanisms and Robustness

6.1 Mechanisms

Two mechanisms could explain these results: **genuine leakage**, where protection pushes clearing activity into unprotected land just across the border, so a nearby "control" cell is itself affected by treatment; and **frontier selection**, where protected areas simply tend to be placed at the edge of accessibility frontiers (Andam et al., 2008; Joppa & Pfaff, 2009), so cells near any PA border would already look different from far-away cells even if protection had no effect at all. Due to these reasons, the main specification deliberately uses a 35km buffer to prevent leakage and additional covariates that can be useful for adjusting for frontier selection.

6.2 Bandwidth Sensitivity

The specifications above hold the control definition fixed. This section asks a different question: does the estimated effect depend on how far from a PA border the comparison group is allowed to sit? Three pieces of evidence help answer this, the first of which was already presented in Section 2.

Where deforestation pressure sits, cross-sectionally. Figure 1 shows that loss rises steadily from a low, flat level deep inside a park to about 1,240 m^2/year right at the edge, then jumps sharply to roughly 1,700 m^2/year just outside, stabilizing across the exterior range, establishing that the PA border is a real structural discontinuity in the landscape, not an arbitrary line.

Whether that pressure contaminates the control pool. Figure 7 plots the empirical semivariogram[4] of pre-treatment forest-loss levels against pairwise distance between grid cells, together with a fitted spherical model (Cressie, 1993). The curve climbs to a local peak of about 12.6 million around 37km, then decreases to about 11.9 million by 53km, and climbs back after 100km, to about 14.1 million by 135km.

A spherical model is additionally fit to gain a more solid understanding of the main buffer used in the analysis. However, this model can only describe a curve that rises once and then settles at a flat level, which is why the fit is restricted to a lag of 40km, since the semivariance peaks around that point. This model gives three numbers that summarize the rise: a nugget of about 8.92 million (the leftover variation between cells even at very short distances), a sill of about 12.58 million (the level the curve settles at), and a range of about 31.9km (the distance up to which nearby cells still share enough local structure to be correlated), with a good fit ($R^2 = 0.95$). The later decrease and rise of the semivariance could reflect that the different regions of Colombia (Andes, Amazon, Caribbean, Pacific, Orinoquia) simply sit at different average loss levels, not that nearby cells become more or less alike with distance.

The fitted ≈ 32km range, together with the local peak around 37km of the semivariance, is the empirical basis for the defined main buffer of 35km threshold above: below that distance, nearby treated and control cells' outcomes are still measurably correlated. This fact can serve as evidence of leakage that would make a "clean" control definition drawn any tighter than this unreliable.

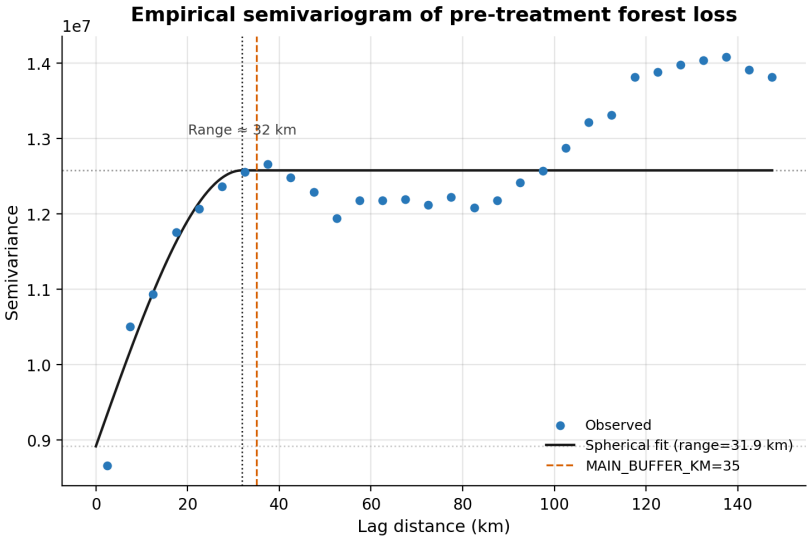


Figure 7: Empirical semivariogram of pre-treatment forest-loss

Figure 8[5] complements this with a distance-indexed, referenced contrast in the spirit of Wang et al. (2020): rather than a single fixed reference band, each 1km bin's pre/post change is benchmarked against a far-exterior reference computed separately per cohort and event time. This reference band was established at $[35, 50]$km as it sits just after the main buffer of 35km.

Unlike the boundary-band grid's single ATT per discrete band, this approach traces the pre/post change in loss **continuously** by distance to the border. The reference band ($[35, 50]$

[4]A (semi)variogram is the key function used to fit a model of the spatial correlation of an observed phenomenon (Bachmaier & Backes, 2011).

[5]Following the design-based, distance-indexed Hajek estimator of Wang et al. (2020): each 1km distance bin's pre/post change is referenced against the far-exterior band's own pre/post change, computed separately per cohort and event time, compensating for shocks shared with that reference.

km) sits at essentially zero, as it must, confirming the far-exterior baseline is well-behaved. Additionally, the curve outside the border converges to zero by roughly 10–15km and stays flat from there, meaning whatever is happening near the border has decayed well before the 35km buffer, corroborating the conclusion the semivariogram reached from raw spatial correlation.

However, it is important to highlight that the convergence to zero just outside the border is not monotonic: there is a positive jump just outside the border, then a slightly negative dip, before converging to zero at around 20km. Additionally, there is more noise inside the border, with more variance and wider confidence intervals. This figure does not resolve leakage versus frontier selection, but confirms that the near-border zone differs from the far reference on both sides and that this difference fully decays before 35km. This approach is mostly descriptive by design, so it serves as a diagnostic alongside the boundary-band grid rather than an additional headline estimate.

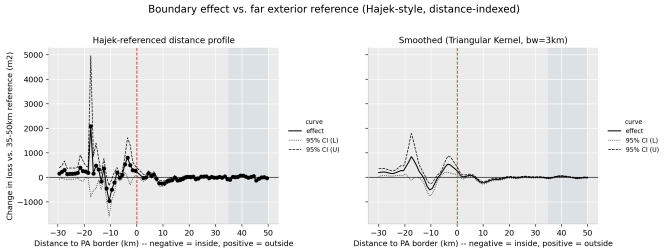


Figure 8: Boundary effect vs. far exterior reference(Hajek-style, distance-indexed)

Whether it actually moves the ATT. Table 6 reports the boundary-band robustness grid directly with no covariates or anticipation, as a sanity check. Figure 4 and Figure 3 (in Section 5) show the same grid extended to all four specifications, overall and by event time respectively. The pattern trends in the expected direction without being perfectly monotonic: as the control band moves farther from the border, the estimated effect goes from a small, only marginally significant positive gap (2–10km: +58.6, $p = 0.093$) to an increasingly large and significant negative effect (10–25km: −135.2, $p < 0.001$), before attenuating again at 25–35km (−81.1, n.s.). This band has the smallest control pool of the six bands, which widens the SE enough to lose significance. The main >35km buffer continues the negative trend at −178.3 (Table 4).

Control band	n control cells	ATT	SE (spatial-HAC)	Sig.
2–10km	218,251	+58.6	34.9	.
2–15km	318,954	+11.8	26.5	n.s.
5–15km	227,652	−26.4	27.2	n.s.
5–25km	366,222	−87.5	26.1	***
10–25km	239,273	−135.2	26.7	***
25–35km	100,153	−81.1	79.8	n.s.

Table 6: Overall ATT at the baseline specification by control band, spatial-HAC SE

By reading the different pieces of evidence together, it is possible to see that the PA border is a genuine structural discontinuity, that spatial correlation in the estimator's own error extends to a fitted range of roughly 35km before stabilizing, and that contamination is large enough to materially shift the estimated ATT, depending on where the control band is drawn. This evidence shows that control cells near the border behave differently, but not why (Section 6).

6.3 Robustness Checks

- The boundary-band control grid above (Table 6) is itself the main robustness check in the paper, run across all four specifications (Figure 4, Figure 3).
- Imputation choices for the doubly-robust covariates (VIIRS regression-imputation, coca zero-fill with a missingness flag; Section 3) are design decisions rather than free parameters, documented so they can be revisited.
- The doubly-robust plug-in's regularization and trimming (Section 4) materially stabilizes both the point estimates and standard errors relative to an earlier, unregularized version of the same estimator, and should be reported here once that comparison is finalized for the paper.

6.4 Threats to Interpretation

The boundary-band grid is a leakage/frontier-selection **diagnostic**, not a competing preferred estimate: a near-border comparison that flips sign once adjusted for covariates is itself evidence against trusting it, which is why the main > 35km buffer carries the paper's headline claim. Two further caveats apply throughout. First, the panel has no new treatment cohort in 2015–2017 or 2022–2023, so cohort-level results in those windows reflect gaps in PA designation activity, not missing data. Second, a single cohort (2013) accounts for the largest cohort-event outlier in the entire dataset, externally corroborated by an independently reported 2016 fire event (Section 5.4); the cohort-size weighting used throughout already down-weights it appropriately, but any reader re-aggregating the underlying event-time estimates without that weighting should be aware of it.

7 Conclusion

Placeholder. Suggested content, now that the design is finalized (Section 5, Section 6):

- **Restate question and answer.** Protected-area designation in Colombia is associated with a robust reduction in annual forest loss, on the order of 14–50% of the control-group mean depending on specification (Table 4), estimated against a control group drawn far enough from any PA border to clear the measured spatial-correlation range (Section 6.2).
- **Main empirical takeaway.** The sign and magnitude of the estimated effect depend materially on where the control group is drawn: near-border bands can flip sign once covariates are added (Figure 4), which is read here as a diagnostic warning rather than a competing estimate.
- **Policy relevance.** If worth making, the policy point likely concerns PA border enforcement/monitoring and the risk that near-border impact evaluations (of this or other PAs) are especially sensitive to control-group choice.
- **Main limitation.** The design cannot fully separate genuine leakage from frontier-selection siting as the source of the near-border gradient (Section 6); the pre-trend evidence is supportive but not a formal test of parallel trends.
- **Next research step.** The land-titling comparison originally scoped for this project (Afro, Indigenous, and Campesino titling) is a natural extension once this PA-only design is finalized, applying the same boundary-band and spatial-HAC machinery to a different treatment definition.

8 Appendix

8.1 Notation Summary

Symbol	Meaning
i	grid cell (1km × 1km)
t	calendar year, 2001, …, 2025
g	cell i's treatment cohort (first PA-designation year), 0 if never treated
Y_{it}	annual forest loss (m^2), cell i, year t
$\text{ATT}(g,t)$	group-time average treatment effect, cohort g at calendar time t
ψ_i	per-unit influence-function contribution feeding the spatial-HAC variance
d_{ij}	distance between cells i and j

Table 7: Notation used throughout the paper.

8.2 Data Construction Details

WDPA snapshot selection. PA boundaries and designation years are pulled as yearly snapshots from the World Database on Protected Areas (UNEP-WCMC and IUCN, 2024), so a cell's recorded treatment cohort reflects the designation year on record at that snapshot rather than the present-day boundary.

Doubly-robust propensity-score fitting. The doubly-robust specifications fit one standardized, cross-validated, L_2-regularized logistic propensity score per cohort (not per group-time cell), with scores trimmed to $[0.02, 0.98]$ before forming the inverse-probability weights. This replaced an earlier, unregularized, per-group-time-cell logistic fit, which produced unstable estimates in specific cohort/control-band combinations; coca-cultivation intensity is genuinely observed for only about 12% of cells, and that sparse, regionally clustered coverage created near-separation in the small per-cell samples the earlier fit used.

Spatial-HAC computation. The variance correction in Section 4 is computed via a streamed KD-tree neighbor search, so the full $N \times N$ pairwise-distance matrix is never materialized; this is the same computational approach used for the empirical semivariogram in Section 6.2.

8.3 Additional Figures and Tables

Candidates to add here once finalized: the unregularized-vs-regularized doubly-robust comparison referenced in Section 6, additional boundary-band maps, and any cohort-level diagnostics beyond cohort 2013 that emerge from further robustness work.

References

Andam, K. S., Ferraro, P. J., Pfaff, A., Sanchez-Azofeifa, G. A., & Robalino, J. A. (2008). Measuring the Effectiveness of Protected-Area Networks in Reducing Deforestation. *Proceedings of the National Academy of Sciences*, *105*(42), 16089–16094. https://doi.org/10.1073/pnas.0800437105

Bachmaier, M., & Backes, M. (2011). Variogram or Semivariogram? Variance or Semivariance? Allan Variance or Introducing a New Term?. *Mathematical Geosciences*, *43*(6), 735–740. https://doi.org/10.1007/s11004-011-9348-3

Baugh, K., Elvidge, C. D., Ghosh, T., & Ziskin, D. (2010). Development of a 2009 Stable Lights Product using DMSP-OLS Data. *Proceedings of the Asia-Pacific Advanced Network*, *30*, 114.

Callaway, B., & Sant'Anna, P. H. C. (2021). Difference-in-Differences with Multiple Time Periods. *Journal of Econometrics*, *225*(2), 200–230.

Callaway, B., Sant'Anna, P. H. C., Quispe, A., & Guevara, C. (2024,). *csdid: Difference-in-Differences with Multiple Time Periods in Python*. https://github.com/d2cml-ai/csdid

Conley, T. G. (1999). GMM Estimation with Cross Sectional Dependence. *Journal of Econometrics*, *92*(1), 1–45.

Cressie, N. A. C. (1993). *Statistics for Spatial Data* (Revised). Wiley-Interscience.

Elvidge, C. D., Baugh, K., Zhizhin, M., Hsu, F.-C., & Ghosh, T. (2017). VIIRS Night-Time Lights. *International Journal of Remote Sensing*, *38*(21), 5860–5879. https://doi.org/10.1080/01431161.2017.1342050

Farr, T. G., Rosen, P. A., Caro, E., Crippen, R., Duren, R., Hensley, S., Kobrick, M., Paller, M., Rodriguez, E., Roth, L., Seal, D., Shaffer, S., Shimada, J., Umland, J., Werner, M., Oskin, M., Burbank, D., & Alsdorf, D. (2007). The Shuttle Radar Topography Mission. *Reviews of Geophysics*, *45*(2), RG2004. https://doi.org/10.1029/2005RG000183

Hansen, M. C., Potapov, P. V., Moore, R., Hancher, M., Turubanova, S. A., Tyukavina, A., Thau, D., Stehman, S. V., Goetz, S. J., Loveland, T. R., Kommareddy, A., Egorov, A., Chini, L., Justice, C. O., & Townshend, J. R. G. (2013). High-Resolution Global Maps of 21st-Century Forest Cover Change. *Science*, *342*(6160), 850–853. https://doi.org/10.1126/science.1244693

Joppa, L. N., & Pfaff, A. (2009). High and Far: Biases in the Location of Protected Areas. *Plos ONE*, *4*(12), e8273. https://doi.org/10.1371/journal.pone.0008273

Marín Llanes, L., Fernández Sierra, M., Vélez, M. A., Martínez-González, E. F., & Murillo-Sandoval, P. J. (2024). *Coca-Based Local Growth and Its Socio-Economic Impact in Colombia* (Documentos CEDE No. 21186). https://repositorio.uniandes.edu.co/entities/publication/21535338-eab8-4145-934c-a7e54b6c2fd9

Revista Semana. (2017,). *Deforestación en Colombia aumentó un 44% entre 2015 y 2016*. Semana. https://www.semana.com/medio-ambiente/articulo/deforestacion-en-colombia-2016-la-perdida-mas-grande-de-los-ultimos-25-anos/38156/

Sant'Anna, P. H. C., & Zhao, J. (2020). Doubly Robust Difference-in-Differences Estimators. *Journal of Econometrics*, *219*(1), 101–122.

UNEP-WCMC and IUCN. (2024,). *Protected Planet: The World Database on Protected Areas (WDPA)*. UNEP-WCMC, IUCN. https://www.protectedplanet.net/

UNODC – Sistema Integrado de Monitoreo de Cultivos Ilícitos (SIMCI). (2023,). *Monitoreo de Territorios Afectados por Cultivos Ilícitos en Colombia*. Oficina de las Naciones Unidas contra la Droga y el Delito.

Wang, Y., Samii, C., Chang, H., & Aronow, P. M. (2020). Design-Based Inference for Spatial Experiments under Unknown Interference. *Arxiv Preprint Arxiv:2010.13599*.